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A Review of Image Annotation Tools for Object Detection

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***Abstract*—With constant research in the last decade, Object Detection has become one of the rapidly evolving sub-fields of Deep Learning. As a result, the complexity and applications of Object Detection models have grown, necessitating larger datasets and multi-format labelled annotations for training and testing. To aid with the annotation work, a variety of tools based on various technologies, techniques, and features have been developed. However, some of the tools only cater to specific standards, requirements and problem domain. Researchers have observed that a drop in the quality of labelled annotations can have unintended consequences for model performance. This study provides an overview of various annotation tools with the goal of assisting researchers in determining the appropriate tool for their needs. The work will also serve as a reference for developers as they work on subsequent features to improve or create new tools.**

Keywords—Deep Learning, Object Detection, Multi-format Labelled Annotation, Image Annotation Tools

I. INTRODUCTION

Deep learning has gotten a lot of attention in the last decade and has become a dominant technology in the field of artificial intelligence. Object detection is regarded as one of the most important areas of deep learning and computer vision. It has been determined in a variety of computer vision applications such as object tracking and retrieval, video surveillance, image captioning, image segmentation, medical imaging, and many others.

Just like any other software project development, a typical object detection project lifecycle goes through various stages- Data Collection, Data Annotation, Model Training and Validation, and Deployment, as shown in Fig. 1.

Supervised machine learning models require data labelling to work effectively. Image annotation is a subset of data labelling where the labelling process focuses only on visual digital data such as images and videos. It often requires manual work. An engineer determines the labels or tags and passes the image-specific information to the computer vision model being trained. Deciding who should perform this manual task is an important strategic decision for the research or project team in the organization. It is because the main methods, namely in-house, outsourcing and crowdsourcing, offer different levels of cost, output quality, data security, etc. [13]. Time required for annotations is lowest in case of crowdsourcing and increases for outsource method and is slowest in case of in-house. It is important to note that there is no prescribed strategy for choosing between these methods. The optimal strategy will vary depending on the conditions and needs of your organization.

TABLE I. Comparison of various Annotation Strategies [13]

Property	Outsource	In-House	Crowdsource
Time Required	Average	High	Low
Price	Average	Expensive	Cheap
Quality of Labelling	High	High	Low
Security	Average	High	Low

With the increase in complexity of use cases that are to be dealt with, the quantity of data is also continuously mounting and becoming very large in aspects like volume and variety. These datasets can contain images with multiple instances of objects within it and thus proper care should be taken while annotating them. The annotation format required can be quite varied and depends on the deep learning model being used, e.g.: YOLO model family uses normalized annotation in a single TXT file per image while Keras RetinaNet uses a single CSV file for the entire dataset where each line represents one bounding box.

Significant attention and portion of time of the Object Detection project lifecycle should be spent in Image Annotation as annotation quality is critical in determining the model performance [1]. It has been evident that patterns are found mainly in coherent data and thus the concept of “garbage in, garbage out” applies [4]. Considering real time projects which might contain many classes and objects in a single image or frame, creating good quality annotations becomes a difficult task and might lead to some errors such as mislabeling the objects, not able to create tightly packed bounding boxes or even positioning them at wrong positions.

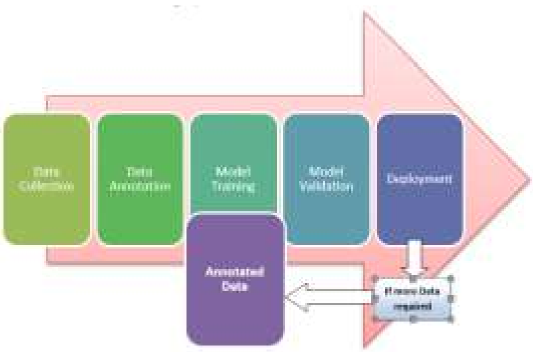

Fig. 1. Object Detection Project Lifecycle

Some of the favorable functionalities of an object detection annotation tool include a simplified User Interface with an option for drawing different annotation shapes like rectangular bounding box, circle, polygon, polylines, etc. Ability to review the annotations is also a fundamental feature which should be present in an annotation tool. Other functionalities like online support, periodical maintenance, collaborative working and option to export annotations in various formats enhances the user experience in all. There are a lot of tools available for the purpose of image annotation, each with its own set of features and different levels of user interaction. Functionalities might vary based on the fact whether the tool is proprietary or open-source. Proprietary tools offer different levels of functionalities based on their pricing plans while most open-source tools are free and offer every feature they have to provide.

The remaining part of the paper is divided into following sections: Section II provides the detailed review for different tools used for image annotation followed by Section III which will describe in brief the proposed work which will lastly be followed by Section IV concluding the paper with some observations and future scope in this domain.

II. RELATED WORK

Experiments were conducted to explore the implications of mislabeled annotations on the model's performance, and [1] reported important conclusions. The authors claim that while open-source labelled datasets such as Microsoft Common Objects in Context (COCO) are accessible, they may not always contain relevant object classes. While collecting and labeling specialized data in-house can be expensive, it gives precisely annotated data, unlike crowd-sourcing methods. Another issue is the amount of the dataset, as complicated models are more data-intensive by nature.

Their experiments try to quantify the trade-off between dataset size and the annotation quality on the model performance. They create a dataset of drones (instance of small object) and create variations of the dataset by introducing degradation in the form of mislabeled annotations. Separate YOLOv4 models were trained on each variation of the dataset and results were observed for various metrics of comparison. The findings from the results indeed confirm that model performance degrades significantly when even a small portion (5%) of the dataset is completely mislabeled. However, larger datasets could be used to compensate the poor annotations to boost the model performance. Their study can further be expanded to see if the results hold up for datasets that contain large objects and multiple classes. Thus, it is quite easy to say that whatever trade-off is selected, be it annotation quality or larger dataset size, a well-designed annotation tool with favorable functionalities must be selected to save time and complete the task with ease.

The authors of [2] created an online tool based on Python and Django, called ImageTagger, to assist teams participating in various RoboCup competitions in performing object detection labelling at a faster pace using collaboration. They conclude that an offline tool would lead to problems with setup and compatibility, so they create an online tool. Functionalities given by the browser-based application include: Manual and Automated Labeling (with filtering), Label Verification, Image Management,

Fig. 2. Annotation View of ImageTagger [2]

Collaboration. They employ their own annotation format for the internal system which could then be exported to a user-defined format. Users can also use the tool to indicate image annotations that are blurry or concealed. They have created the user interface, data and user management systems in such a way that the entire process runs smoothly.

In 2018, Matthieu Pizenberg and collaborators created a web-based open-source application for creating customizable annotations [3]. The server is written in Node.js and is kept as simple as possible, while the frontend is built with Elm. The application loads the images and annotates them locally rather than sending the data to any

```
{ "classes":
  [ { "category": "Person"
    , "classes": [ "person" ]
    }
  , { "category": "Animal"
    , "classes": [ "bird", "cat", "cow", "dog", "horse",
      "sheep" ]
    }
  , { "category": "Vehicle"
    , "classes": [ "aeroplane", "bicycle", "boat", "bus",
      "car", "motorbike", "train" ]
    }
  , { "category": "Indoor"
    , "classes": [ "bottle", "chair", "dining table", "
      potted plant", "sofa", "tv/monitor" ]
    }
  ]
, "annotations": [ "point", "bbox", "stroke", "outline",
  "polygon" ]
}
```

Fig. 3. Configuration file used for specifying Classes and Annotation types in [3]

external server, ensuring that the user's files are kept private and secure. To create ground-truth labelling, it offers five different types of annotation shapes: bbox, polygon, point, stroke, and outline. Using a custom JSON configuration file, the five main shaping techniques can then be employed to create variations. The classes and their categories are specified in the same configuration file. The application also works with the Mechanical Turk to provide crowdsourcing for annotation creation and management.

[4] developed an open-source annotation software which allows for customized preferences while also generating statistical data reports. It uses Python GUI framework called Qt5 for a user-friendly visual interface and uses MongoDB for database storage. The tool can also be customized by the user for efficient handling. It also works with several different types of inputs like images, audio, general signals, etc. User can also add extra information called properties

Fig. 4. Tool created by [4] being used to annotate a traffic dataset

related to the labelled image in context of the dataset. The tool is also capable of semi-automating the annotation task by using pre-trained algorithms which could further be perfected under the supervision of human expert. Annotations can be exported in different formats, including TXT and CSV files. The software deals most of its logic using database files and thus allows it to provide user the option of collaborative work.

In order to perform annotations easier, faster and at higher quality, authors of [5] created first ever general-purpose, browser-based extension for image annotation named BRIMA. The tool offers easy and comfortable UI experience so as to allow users to perform annotations with high accuracy. It uses the browser-viewport to annotate the image using polygon annotation. The features of the tool include minimal client-side and server-side setup, support for different combinations of OS and browser with no pre-built dataset requirement. The tool is written in pure JavaScript with server-side implementations in PHP and Python using Flask to help researchers and crowdsource contributors. It also has support for keyboard shortcuts which can be customized according to user's choice. The only downside is that the tool natively exports the annotations only in MS COCO JSON format. To test the effectiveness of BRIMA, the tool was validated by various crowdsource annotators. Statistics of their work is mentioned in Table II.

TABLE II. Validation Statistics for BRIMA [5]

Annotator	**Approved Images**	**OS and Browser**	**Easy Setup and Use (0-5)**	**Overall Experience (0-5)**
Person1 JU	418	MacOS (10.15.6) FFox (80.0.1)	3.5	2
Person2 PK	525	Debian (9.13) Chromium (73.0.3683.75-dev)	5	4
Person3 SK	542	Win10 FFox (81.0.1)	5	4
Person4 AA	228	no response	no response	no response
Person5 AT	632	Win10 (1809) FFox (78.3.0esr)	3	4
Person6 TL	750	Win10 FFox (80.0.1)	5	4
Person7 LS	977	Ubuntu (18.04.5 LTS) Chrome (85.0.4183.121)	5	5
Person8 AC	95	Ubuntu (16.04 LTS) FFox (68.7.0esr) 32-bit	5	5
avg/**sum**	520.87/**4167**	-	4.5/-	4/-

Researchers at the Visual Geometry Group developed VGG Image Annotator (VIA) Tool [6], an open-source tool which helps in creating ground truth labels for multiple types of input data. It is a browser-based tool and is developed solely using the technologies of HTML, JavaScript and CSS. Minimalist approach of the interface makes it intuitive and user-friendly. The developers have allowed the VIA tool to be easily configurable, thus extending its use cases. It provides multiple options to annotate the spatial region of the image: rectangle, circle, polygon, polyline and freehand drawn mask among others.

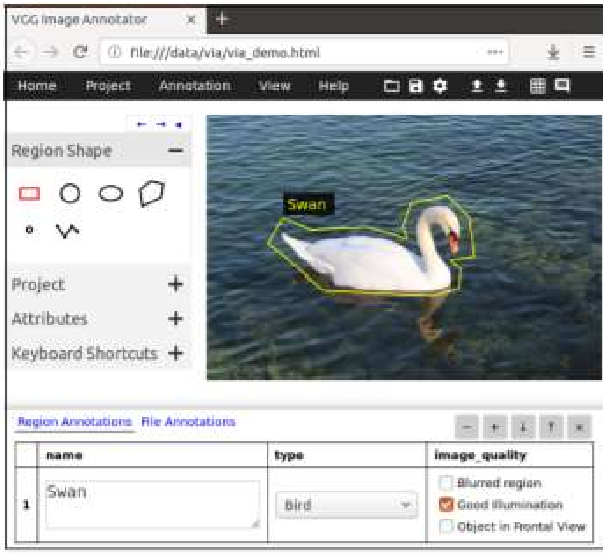

Fig. 5. Interface of the VGG Image Annotator [6]

Additionally, textual description of each annotation can be provided for which options like checkbox and dropdown are available. It also provides a set of hotkey shortcuts for faster use. These ground-truth annotations can be exported to plain text data formats like JSON and CSV, allowing them to be processed and trained by other software tools. VIA also allows a group of human annotators to collaborate on the annotation of a huge dataset. However, it lacks the dataset management functionality.

Another widely used object detection graphical image annotation software is LabelImg [7], which installs locally on the user's machine. The visual interface is built with the Qt toolkit and is written in Python. PASCAL VOC (XML), YOLO, and CreateML annotation formats are all supported. The software is available for Windows, macOS, and Linux. Shortcut hotkeys are also supported by the tool, allowing you to complete the annotation work more quickly. The application also has a feature that allows you to flag images for verification. The tool is simple to use, but it lacks some advanced features that other platforms offer, such as collaboration and management preferences.

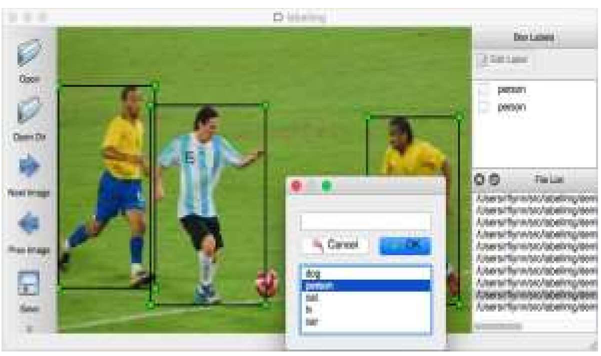

Fig. 6. Interface of LabelImg [7]

There are tools that are similar to LabelImg, such as Labelme [8], which is based on a tool with the same name (developed by the MIT Computer Science and Artificial Intelligence Laboratory) [9], and provides advanced options for annotating images for tasks such as object detection, semantic and instance segmentation, classification, and video annotation.

Additionally, we also encounter some segmentation annotation tools which are studied and developed in [10], [11], [12].

Compared to the previously discussed papers which were created for more generic purpose of image annotations, Jorge Bernal with other developers in [10] presented, GTCreator, a flexible annotation tool for creating image and text annotations in the healthcare domain, with a focus on polyp datasets in colonoscopy. The tool retains the fundamental functionality of other similar tools while adding various capabilities (such as allowing many annotators to work on the same task at the same time and allowing for easy annotation transfer) to target enormous datasets. The tool offers various functionalities like creating image and text-based annotations, review process by inspection of the marked images using filtering capabilities, easy experimental

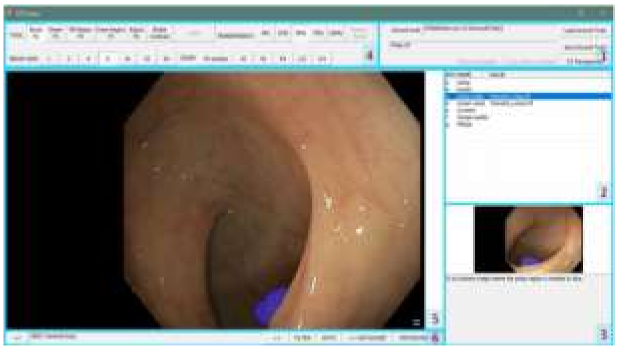

Fig. 7. Interface of GTCreator [10]

setup by providing splitting of the annotated CSV data using text-based metadata set by the user. All of these characteristics contribute to the creation of quick and precise annotations. Additional tools in the medical domain include [11] where authors create a semi-automatic annotation tool using Deep Learning model, U-Net for faster and accurate annotations of cellular medical images like nucleus and cytoplasm. It supports bboxannotations.

ByLabel [12] is a semi-automatic boundary-based annotation tool developed with an aim to reduce the number of image-clicks and improve the annotation accuracy. It uses edge detection and splitting algorithms to detect boundary of the target objects for annotation. The user can then select a subset of these detected edges to build the object borders by concatenating them with the original image. The tool also provides an option for the user to draw the boundaries to correct errors created by automatic edge detection. This method reduces the number of image clicks and annotation time by 73% and 56%, respectively, while also improving annotation accuracy.

Above mentioned tools were mostly developed as open-source tools or part of a research team's project. We would also like to mention some proprietary tools like Labelbox [14] and roboflow [15] which are full-fledged and provide many advance functionalities (including collaboration, automation, data and user management, support for multiple formats, etc.) which assist in easier creation of annotations. They have their pricing plans available on their respective websites.

III. PROPOSED METHODOLOGY

We propose the creation of an annotation tool that

TABLE III. Comparison of various Tools explored in Related Work section

Tool	**Year**	**Annotation Types**	**Export Format**	**FOSS/ Paid**	**Collaborative**	**Locally Hosted**	**Extra Features**
ImageTagger [2]	2018	bbox, polygon, line, point	User Definable	FOSS	Yes	No	Label Verification, Automated Annotation, Image Management
[3]	2018	bbox, polygon, point, stroke, outline	JSON	FOSS	No	Online and Offline options available	MTurk Integration, Configurable Interface
[4]	2020	rectangle, ellipse, polygon, etc.	TXT, CSV, etc.	FOSS	Yes	Yes	Semi-automatic annotation, Configurable Interface, multi-domain input, statistical report generation
BRIMA [5]	2021	Polygon	JSON	FOSS	No	Yes	None
VIA [6]	2017	rectangle, circle, polygon, polyline, freehand drawn mask	JSON, CSV	FOSS	Yes	Online and Offline options available	Configurable, provides textual description of the annotations
LabelImg [7]	2015	bbox	Pascal VOC (XML), YOLO (TXT), CreateML	FOSS	No	Yes	Option to flag images for verification, shortcut hotkeys are available
Labelme [8]	2018	Polygon, rectangle, circle, line, point	JSON	FOSS	No	Yes	Option for video annotations, segmentation.
AnnotatorJ [11]	2020	bbox	TXT	FOSS	No	Plugin for ImageJ	Semi-automatic annotations, used in segmentation
LabelBox [14]	2018	Bbox, polygon, line, point, etc.	Multiple formats	Paid	Yes	No	Segmentation, automated annotation, reviewing functionality, multi-domain input, api, end-to-end platform
RoboFlow [15]	2020	bbox	JSON, XML, CSV, TXT, etc.	Paid	Yes	No	End-to-end platform, segmentation, support for multiple image formats and processing, integration with other labelling and modeling tools.

combines the best characteristics of all of the above-mentioned tools while also attempting to mitigate the weaknesses of the various tools. The annotation software will be free and open source. It will have a simple and intuitive user interface that requires little to no training to use and assures a quick and effective workflow for users. Annotations will be able to be exported in a variety of formats, depending on the user's preferences. There may be a large number of classes, but the application may be specific and only require a few classes to deal with, in which case the annotation tool will have a filtering option to swiftly browse through the relevant images. The tool will provide several reviewing functions and flagging options to verify that users' annotations are correct. The tool would be browser-based and set up locally at first. In future versions, the utility will be hosted over the internet and will only require minimum installation. Third-party integrations with various tools will be available. Apart from that, the annotation tool will evolve into an end-to-end platform that will allow model training, validation, and a broader range of annotation activities such as segmentation and classification.

IV. CONCLUSION AND FUTURE WORK

Every process has a beginning point, and for object detection, that starting point is image annotation. We went through a lot of research papers and tools and came up with a few essential points. The most important phase in object detection is annotation, which has a direct impact on model performance. Image annotation is available in a variety of formats. Image annotation can be done using either locally installed or cloud-based applications. There are also several image annotation methodologies, such as in-house, outsource, and crowdsource, which may influence the tool the user chooses. Each has its own set of advantages and disadvantages. Choosing the best strategy is more dependent on the problem domain. Image annotation is reliant on a number of elements, including a properly sized bounding box and properly labelled data, to mention a few. Over the years, interesting research has been conducted in this area, and more attention is now being paid to automating many of the manual processes encountered during image annotation in order to create a more efficient and user-friendly workflow.

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